Forecasting On Air Passenger data

Project Pipeline

Load Data ↓ Clean & Handle Missing Values ↓ Convert Date Index ↓ Visualization ↓ Stationarity Check ↓ Differencing ↓ Train/Test Split ↓ Model Training (ARIMA) ↓ Forecast ↓ Evaluation ↓ Future Prediction

In [118…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from sklearn.metrics import mean_absolute_error, mean_squared_error
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
from sklearn.metrics import mean_absolute_percentage_error
from pmdarima import auto_arima
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

Load AirPassengers Dataset

In [119…

```
df=pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\AirPassengers.csv')
```

In [120…

```
df.head()
```

Out[120…

	Month	Passengers
0	1949-01	112
1	1949-02	118
2	1949-03	132
3	1949-04	129
4	1949-05	121

In [121…

```
df.tail()
```

Out[121…

	Month	Passengers
139	1960-08	606
140	1960-09	508
141	1960-10	461
142	1960-11	390
143	1960-12	432

In [122…

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   Month       144 non-null    object
 1   Passengers  144 non-null    int64 
dtypes: int64(1), object(1)
memory usage: 2.4+ KB
```

Convert Month Column

In [123…
```
df['Month'] = pd.to_datetime(df['Month'])
```

In [124…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Month       144 non-null    datetime64[ns]
 1   Passengers  144 non-null    int64
dtypes: datetime64[ns](1), int64(1)
memory usage: 2.4 KB
```

In [125…
```
df.set_index('Month', inplace=True)
```

Check Missing Values

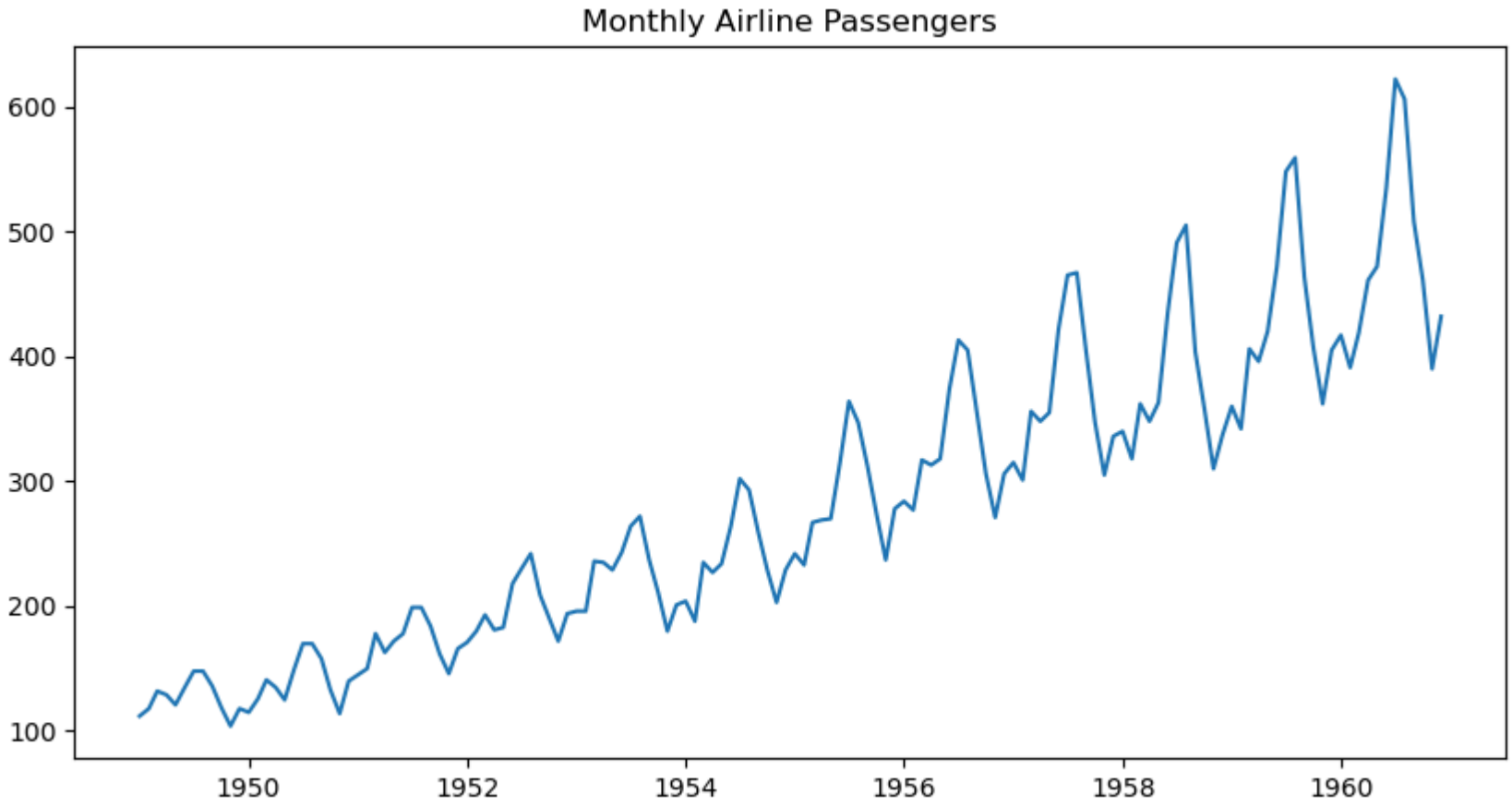
In [126…
```
df.isnull().sum()
```

Out[126…
```
Passengers    0
dtype: int64
```

In [127…
```
df = df.dropna()
```

Visualize Time Series

In [128…
```
plt.figure(figsize=(10,5))
plt.plot(df)
plt.title("Monthly Airline Passengers")
plt.show()
```



Check Stationarity

In [129…

```
result = adfuller(df['Passengers'])
print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: 0.8153688792060511
p-value: 0.991880243437641
```

Interpretation:

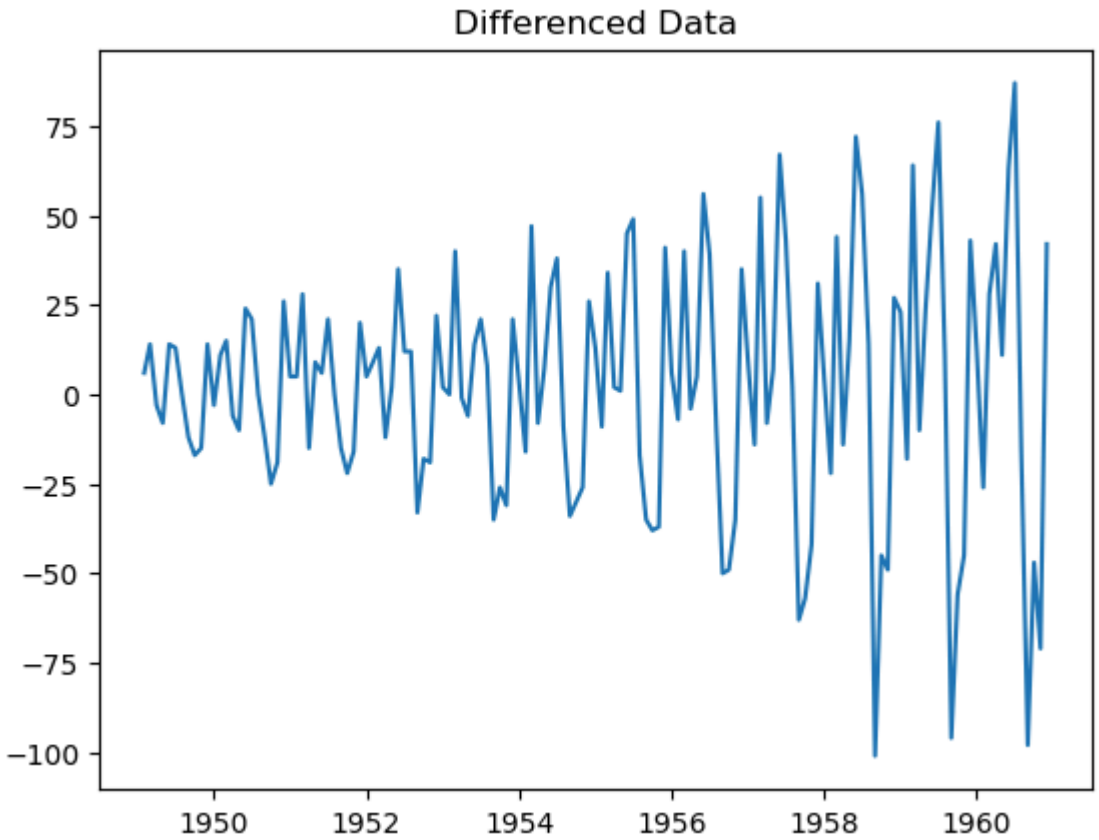
p < 0.05 → Stationary

p > 0.05 → Need differencing

Difference the Series

In [130…

```
df_diff = df.diff().dropna()
plt.plot(df_diff)
plt.title("Differenced Data")
plt.show()
```



Second Stationarity Checking

In [131…

```
result = adfuller(df_diff['Passengers'])
print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

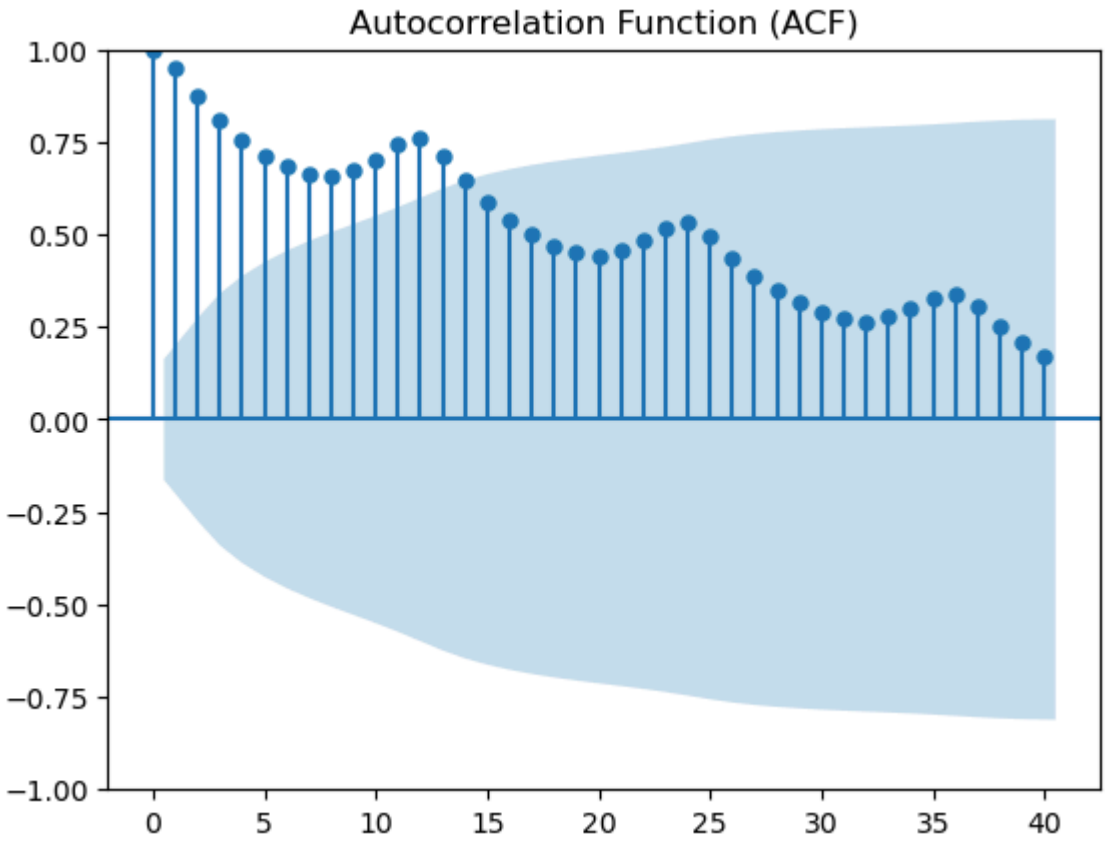
```
ADF Statistic: -2.8292668241699874
p-value: 0.05421329028382711
```

ACF PLot

In [132…

```
plt.figure(figsize=(10,5))
plot_acf(df['Passengers'], lags=40)
plt.title("Autocorrelation Function (ACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

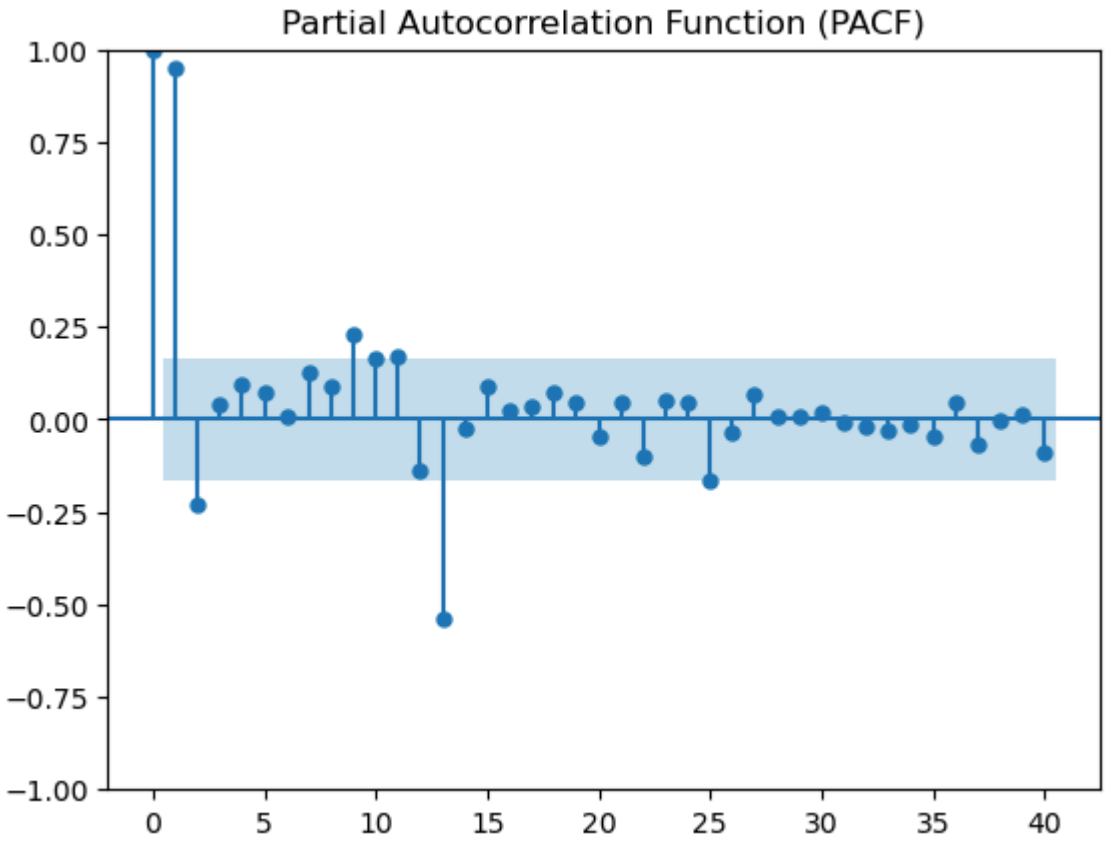


Pacf Plat

In [133…

```
plt.figure(figsize=(10,5))
plot_pacf(df['Passengers'], lags=40)
plt.title("Partial Autocorrelation Function (PACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



Train Test Split

In [134…

```
train_size = int(len(df) * 0.8)
train = df[:train_size]
test = df[train_size:]
```

Build ARIMA Model

In [135…

```
model = ARIMA(train['Passengers'], order=(4,1,12))
model_fit = model.fit()
```

```python
print(model_fit.summary())
```

```
                               SARIMAX Results
==========================================================================
Dep. Variable:             Passengers   No. Observations:                  115
Model:                ARIMA(4, 1, 12)   Log Likelihood                -476.574
Date:                Thu, 12 Mar 2026   AIC                            987.148
Time:                        13:37:53   BIC                           1033.664
Sample:                    01-01-1949   HQIC                          1006.026
                         - 07-01-1958
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          1.6633      0.070     23.864      0.000       1.527       1.800
ar.L2         -1.8088      0.127    -14.231      0.000      -2.058      -1.560
ar.L3          1.5601      0.127     12.307      0.000       1.312       1.809
ar.L4         -0.9266      0.059    -15.829      0.000      -1.041      -0.812
ma.L1         -1.7062      0.607     -2.813      0.005      -2.895      -0.517
ma.L2          1.5794      1.788      0.883      0.377      -1.925       5.084
ma.L3         -1.4709      4.625     -0.318      0.750     -10.535       7.594
ma.L4          0.8814      3.186      0.277      0.782      -5.362       7.125
ma.L5          0.1661      0.647      0.257      0.797      -1.102       1.434
ma.L6         -0.0494      1.169     -0.042      0.966      -2.340       2.242
ma.L7          0.1753      0.729      0.240      0.810      -1.254       1.605
ma.L8         -1.0306      0.996     -1.035      0.301      -2.982       0.921
ma.L9          1.3238      0.971      1.363      0.173      -0.580       3.228
ma.L10        -1.4271      2.684     -0.532      0.595      -6.687       3.833
ma.L11         1.5528      4.238      0.366      0.714      -6.754       9.860
ma.L12        -0.6597      1.700     -0.388      0.698      -3.991       2.672
sigma2       192.6943    489.420      0.394      0.694    -766.551    1151.939
===================================================================================
Ljung-Box (L1) (Q):                   2.95   Jarque-Bera (JB):                 1.90
Prob(Q):                              0.09   Prob(JB):                         0.39
Heteroskedasticity (H):               4.44   Skew:                             0.31
Prob(H) (two-sided):                  0.00   Kurtosis:                         2.92
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Forecast

In [136…

```
forecast = model_fit.forecast(steps=len(test))
forecast = pd.Series(forecast, index=test.index)
forecast
```

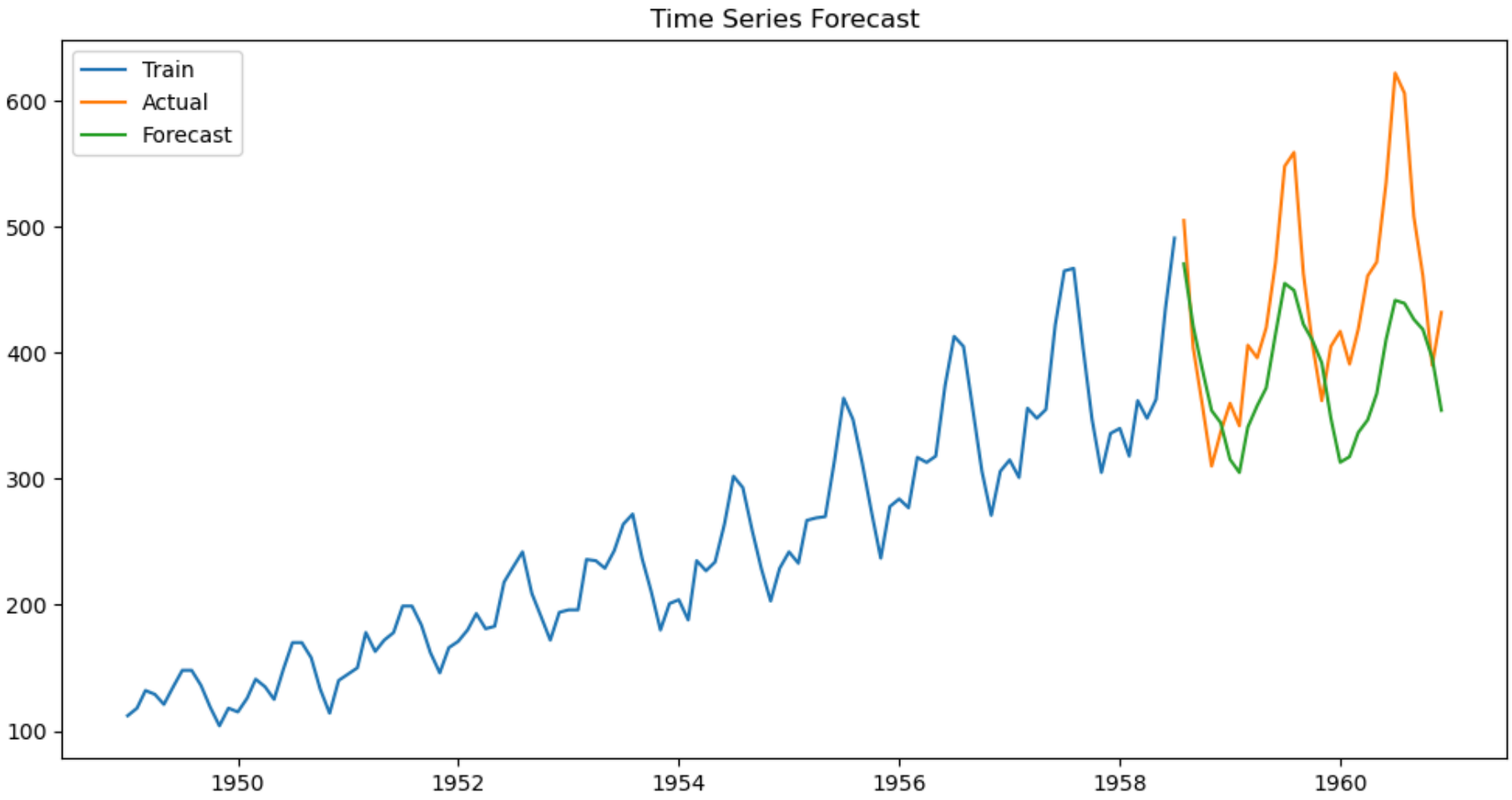
Out[136…

```
Month
1958-08-01    470.543158
1958-09-01    421.028322
1958-10-01    387.025526
1958-11-01    354.167550
1958-12-01    344.483601
1959-01-01    315.376145
1959-02-01    304.999112
1959-03-01    341.036868
1959-04-01    357.994043
1959-05-01    372.194188
1959-06-01    415.348540
1959-07-01    455.073694
1959-08-01    449.531771
1959-09-01    422.625184
1959-10-01    409.883699
1959-11-01    391.905139
1959-12-01    348.207297
1960-01-01    313.098103
1960-02-01    317.500037
1960-03-01    336.814279
1960-04-01    346.694255
1960-05-01    367.590790
1960-06-01    410.529709
1960-07-01    441.669344
1960-08-01    439.241252
1960-09-01    426.502512
1960-10-01    418.499676
1960-11-01    395.589064
1960-12-01    354.334062
Name: predicted_mean, dtype: float64
```

Visualization of Forecast

In [137…

```
plt.figure(figsize=(12,6))
plt.plot(train, label="Train")
plt.plot(test, label="Actual")
plt.plot(forecast, label="Forecast")
plt.legend()
plt.title("Time Series Forecast")
plt.show()
```



Evaluate Model

In [140…

```
mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
y_true = test['Passengers']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
```

```
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 14.153692801273564
MAE: 65.85099595444427
RMSE: 79.31109973905562
```

Future Forecast

Forecast next 30 days.

In [141…

```
future = model_fit.forecast(steps=30)
print(future)
```

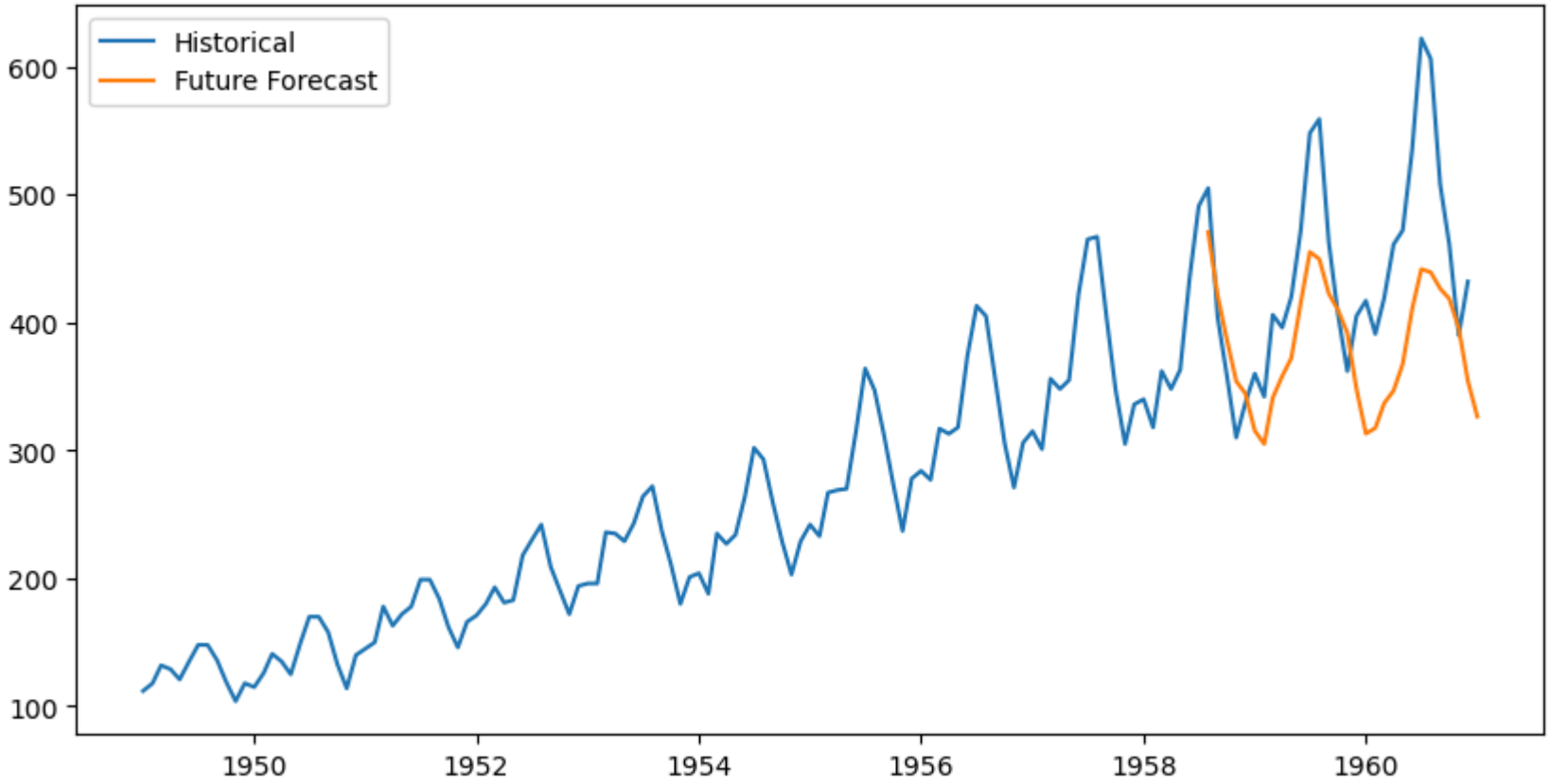
```
1958-08-01    470.543158
1958-09-01    421.028322
1958-10-01    387.025526
1958-11-01    354.167550
1958-12-01    344.483601
1959-01-01    315.376145
1959-02-01    304.999112
1959-03-01    341.036868
1959-04-01    357.994043
1959-05-01    372.194188
1959-06-01    415.348540
1959-07-01    455.073694
1959-08-01    449.531771
1959-09-01    422.625184
1959-10-01    409.883699
1959-11-01    391.905139
1959-12-01    348.207297
1960-01-01    313.098103
1960-02-01    317.500037
1960-03-01    336.814279
1960-04-01    346.694255
1960-05-01    367.590790
1960-06-01    410.529709
1960-07-01    441.669344
1960-08-01    439.241252
1960-09-01    426.502512
1960-10-01    418.499676
1960-11-01    395.589064
1960-12-01    354.334062
1961-01-01    326.474064
Freq: MS, Name: predicted_mean, dtype: float64
```

Plot future prediction:

In [142…

```
plt.figure(figsize=(10,5))
plt.plot(df, label="Historical")
plt.plot(future, label="Future Forecast")
```

```
plt.legend()
plt.show()
```



**********Forecasting On WALLMART data**********

Project Workflow

1 Data Collection (Walmart Dataset) ↓ 2 Data Cleaning & Preprocessing ↓ 3 Exploratory Data Analysis ↓ 4 Feature Aggregation ↓ 5 Train/Test Split ↓ 6 ARIMA Model Training ↓ 7 Forecasting ↓ 8 Model Evaluation ↓ 9 Future Sales Prediction

In [143…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from sklearn.metrics import mean_absolute_error, mean_squared_error
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
from sklearn.metrics import mean_absolute_percentage_error
from pmdarima import auto_arima
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

Load Dataset

In [144…

```
df_1=pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\train.csv')
```

In [145…

```
df_1.head()
```

Out[145…

	Store	Dept	Date	Weekly_Sales	IsHoliday
0	1	1	2010-02-05	24924.50	False
1	1	1	2010-02-12	46039.49	True
2	1	1	2010-02-19	41595.55	False
3	1	1	2010-02-26	19403.54	False
4	1	1	2010-03-05	21827.90	False

Data Preprocessing

Convert Date column to datetime.

In [146…
```
df_1['Date'] = pd.to_datetime(df_1['Date'])
```

In [147…
```
df_1 =df_1.sort_values('Date')
```

In [238…
```
df_1 = df_1[df_1["Weekly_Sales"] > 0]
```

Aggregate Weekly Sales

In [239…
```
sales = df_1.groupby('Date')['Weekly_Sales'].sum().reset_index()
sales.head()
```

Out[239…

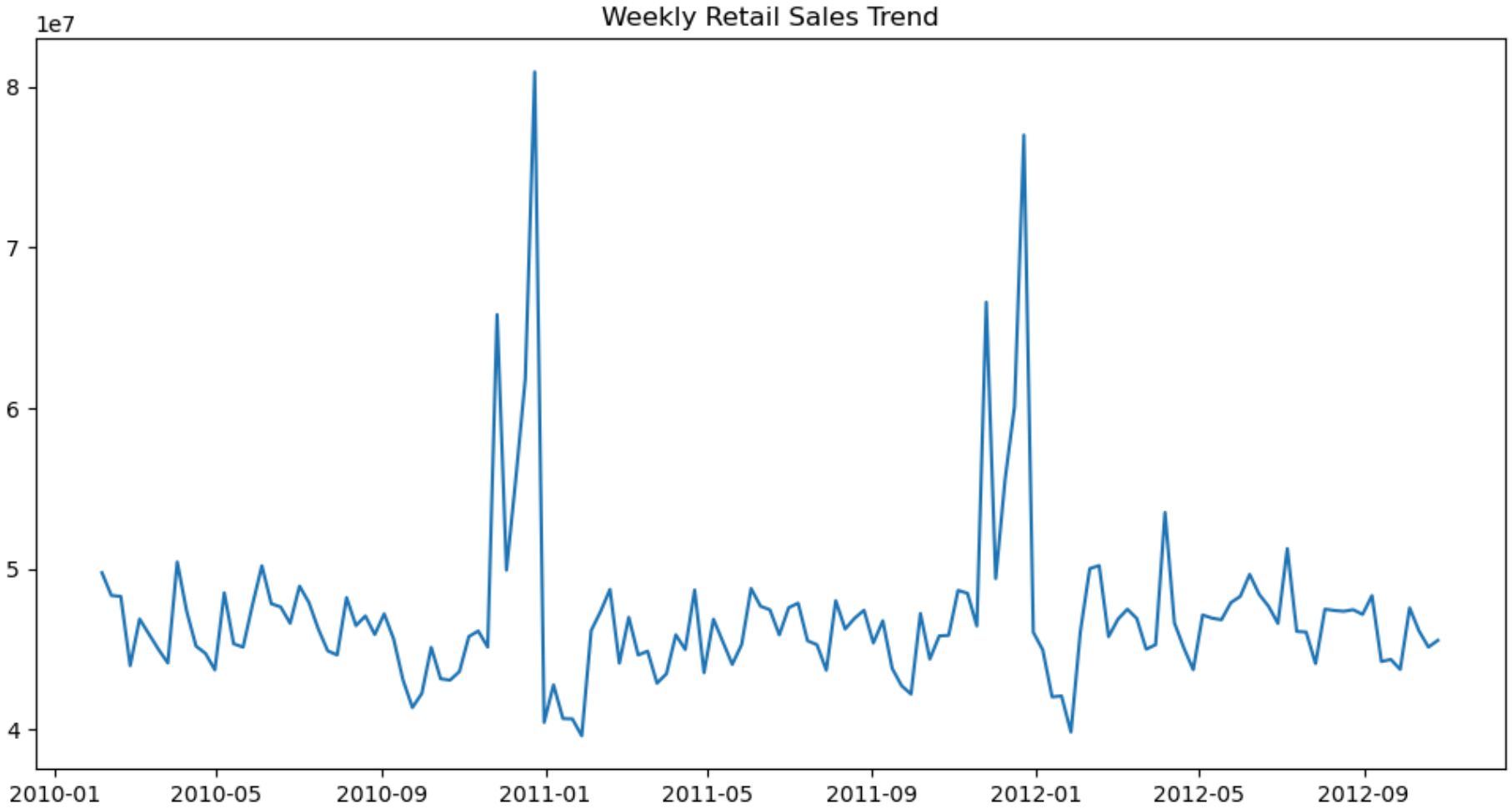
	Date	Weekly_Sales
0	2010-02-05	49750875.98
1	2010-02-12	48336800.10
2	2010-02-19	48277902.33
3	2010-02-26	43970440.65
4	2010-03-05	46872715.16

In [240…
```
sales.set_index('Date', inplace=True)
```

Exploratory Data Analysis

In [241…

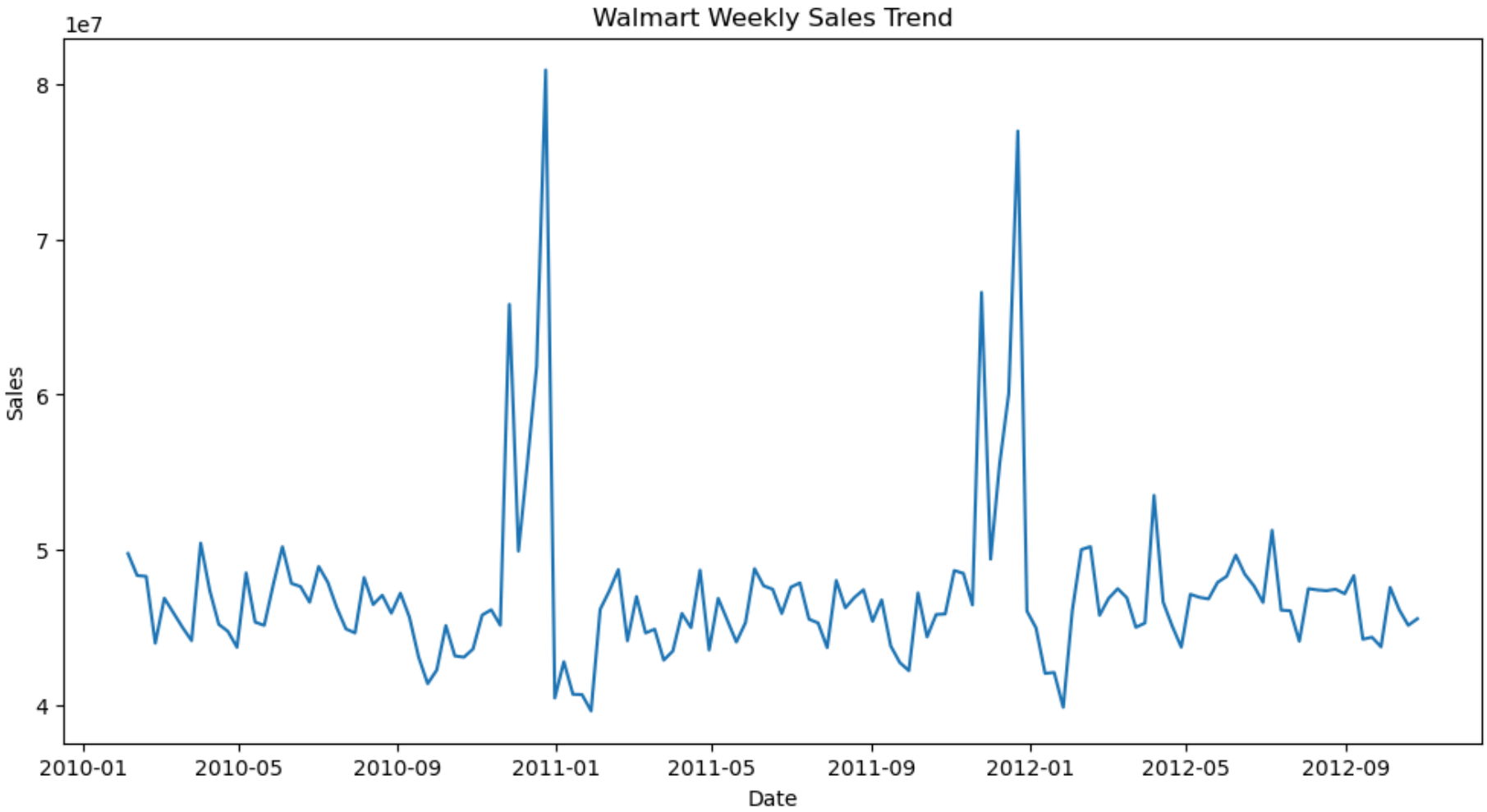
```
plt.figure(figsize=(12,6))
plt.plot(sales)
plt.title("Weekly Retail Sales Trend")
plt.show()
```



In [242…

```
plt.figure(figsize=(12,6))
plt.plot(sales)
plt.title("Walmart Weekly Sales Trend")
plt.xlabel("Date")
```

```
plt.ylabel("Sales")
plt.show()
```



Check Stationarity

In [243…

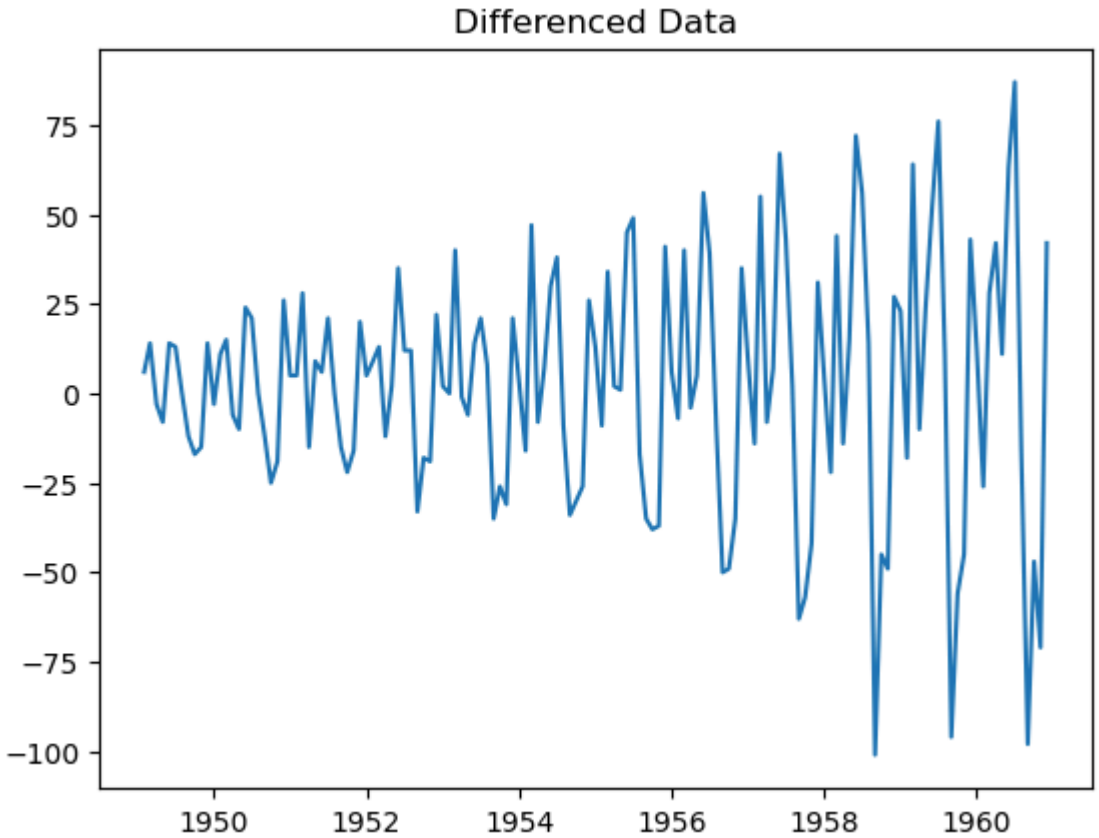
```
result = adfuller(sales['Weekly_Sales'])
print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: -5.9083977177974365
p-value: 2.67460678882577e-07
```

Difference the series

In [244…

```
sales_diff =  sales.diff().dropna()
plt.plot(df_diff)
plt.title("Differenced Data")
plt.show()
```



Second Stationarity Checking

In [245…

```
result = adfuller(sales_diff['Weekly_Sales'])
print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: -6.69952805865287
p-value: 3.9213088895704494e-09
```

Train Test Split

In [246…

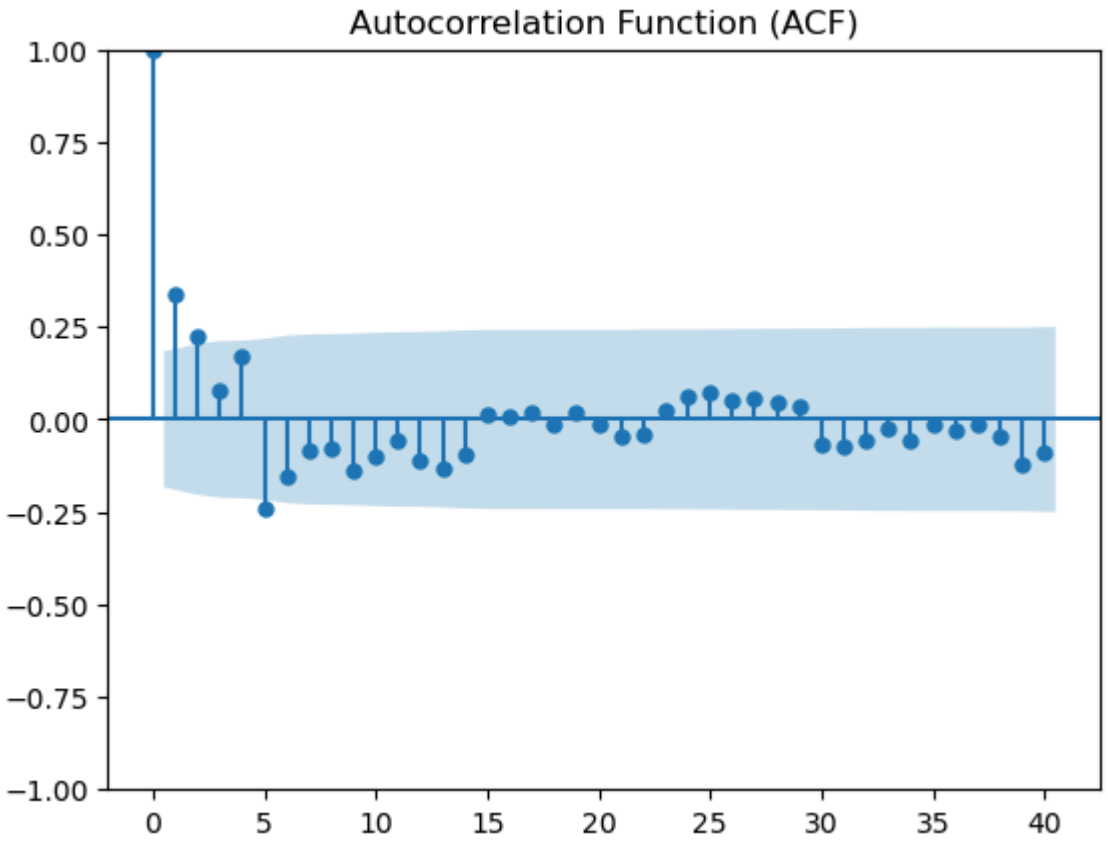
```
train_size = int(len(sales)*0.8)
train = sales[:train_size]
test = sales[train_size:]
```

ACF Plot

In [247…

```
plt.figure(figsize=(10,5))
plot_acf(train['Weekly_Sales'], lags=40)
plt.title("Autocorrelation Function (ACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

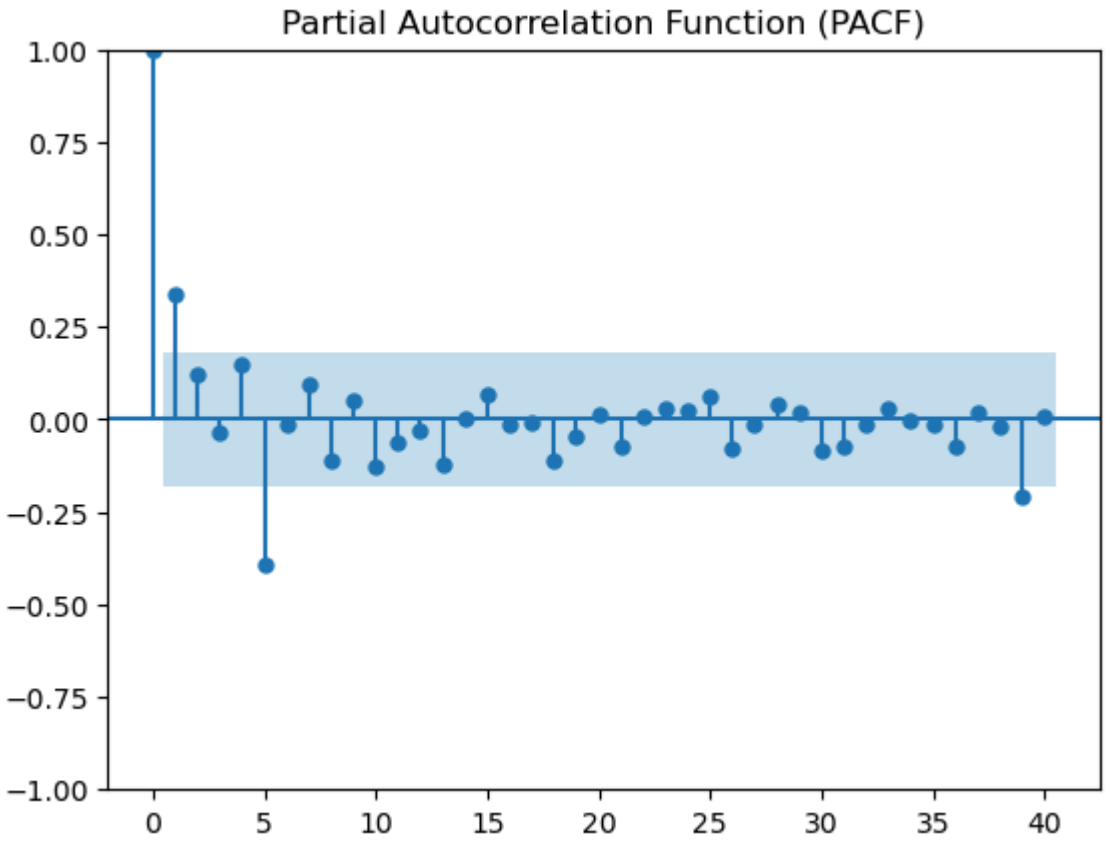


PACF Plot

In [248…

```
plt.figure(figsize=(10,5))
plot_pacf(train['Weekly_Sales'], lags=40)
plt.title("Partial Autocorrelation Function (PACF)")
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



Build ARIMA Model

In [249…

```
model = ARIMA(train['Weekly_Sales'], order=(3,0,2))
model_fit = model.fit()
print(model_fit.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:           Weekly_Sales   No. Observations:                  114
Model:                 ARIMA(3, 0, 2)   Log Likelihood               -1929.168
Date:                Thu, 12 Mar 2026   AIC                           3872.335
Time:                        14:16:13   BIC                           3891.489
Sample:                    02-05-2010   HQIC                          3880.108
                         - 04-06-2012
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       4.722e+07   3.13e+05    151.044      0.000    4.66e+07    4.78e+07
ar.L1          0.5720      0.269      2.127      0.033       0.045       1.099
ar.L2          0.7399      0.302      2.447      0.014       0.147       1.333
ar.L3         -0.4187      0.100     -4.196      0.000      -0.614      -0.223
ma.L1         -0.2855      0.363     -0.786      0.432      -0.998       0.427
ma.L2         -0.7145      0.277     -2.583      0.010      -1.257      -0.172
sigma2      2.776e+13   7.57e-05   3.67e+17      0.000    2.78e+13    2.78e+13
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):               486.00
Prob(Q):                              0.97   Prob(JB):                         0.00
Heteroskedasticity (H):               7.20   Skew:                             2.30
Prob(H) (two-sided):                  0.00   Kurtosis:                        12.01
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
[2] Covariance matrix is singular or near-singular, with condition number 5.91e+34. Standard errors may be unstable.
```

Forecast Test Data

In [250…

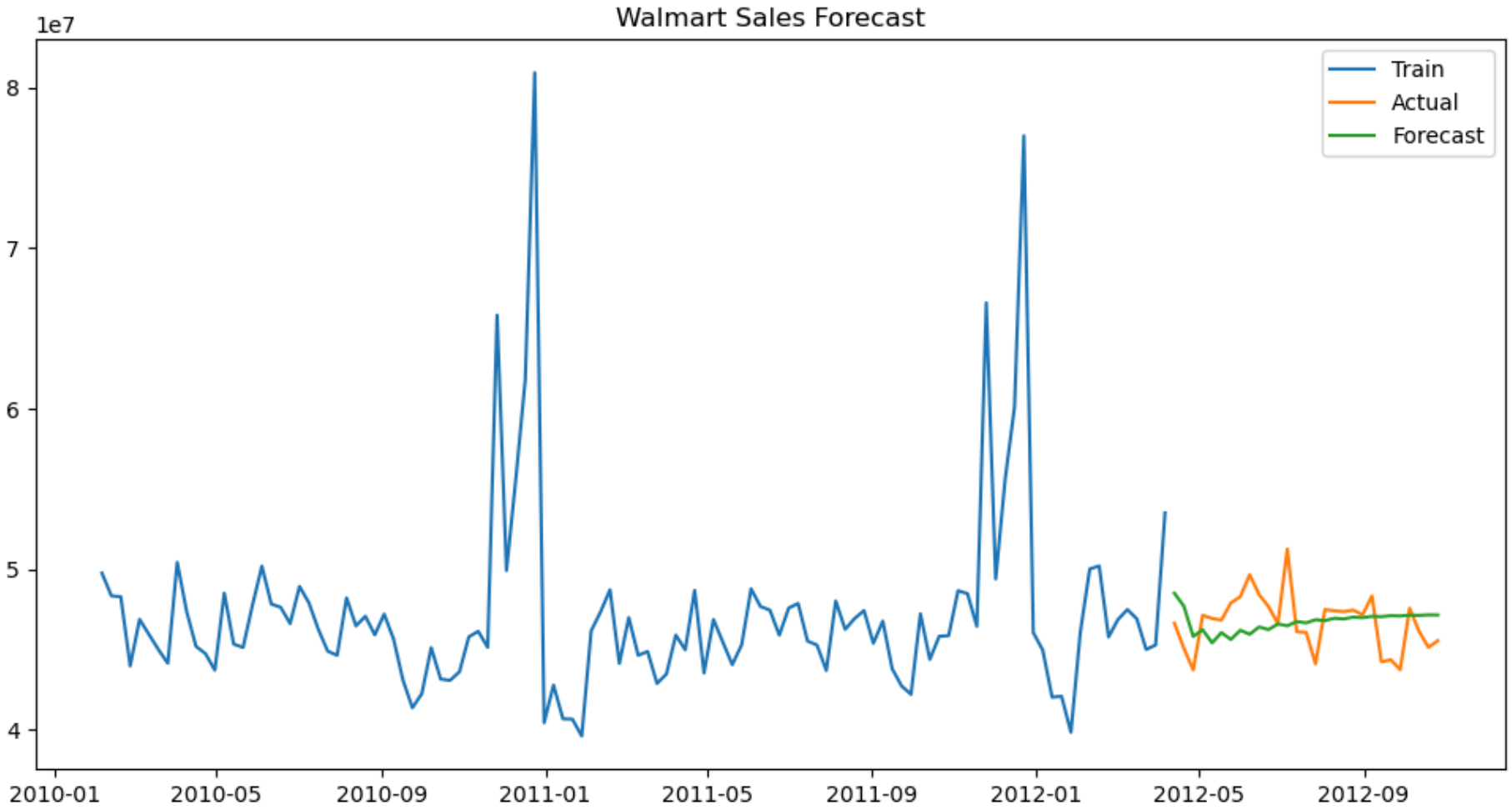
```
forecast = model_fit.forecast(steps=len(test))
forecast = pd.Series(forecast, index=test.index)
print("Test length:", len(test))
print("Forecast length:", len(forecast))
```

```
Test length: 29
Forecast length: 29
```

Visualization

In [251…

```
plt.figure(figsize=(12,6))
plt.plot(train, label="Train")
plt.plot(test, label="Actual")
plt.plot(forecast, label="Forecast")
plt.legend()
plt.title("Walmart Sales Forecast")
plt.show()
```



Model Evaluation

In [252…

```
mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
y_true = test['Weekly_Sales']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
```

```
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 3.5179218968452095
MAE: 1639465.133381563
RMSE: 2002416.6682305136
```

Forecast Future Sales

In [253…

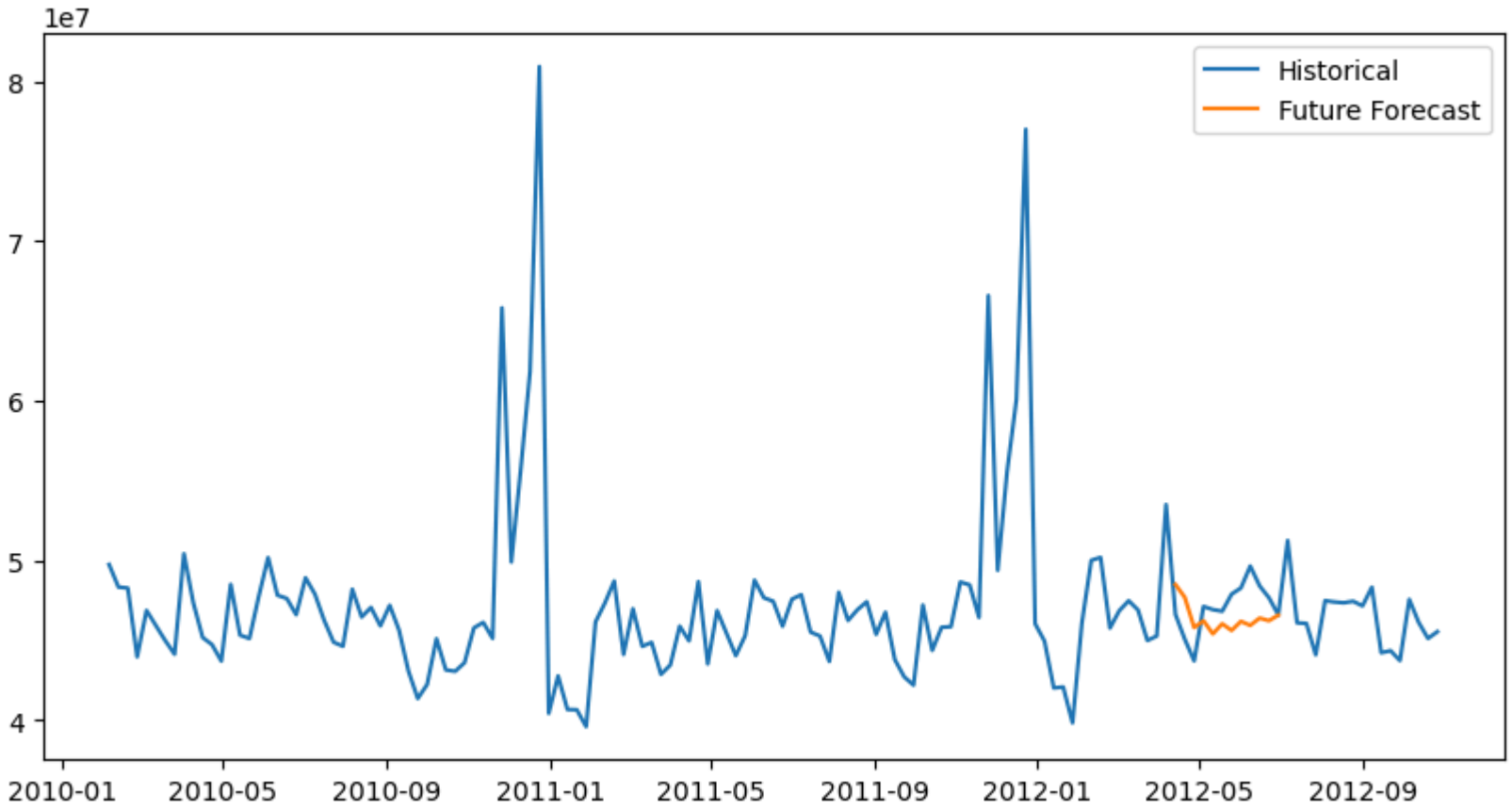
```
future = model_fit.forecast(steps=12)
print(future)
```

```
2012-04-13    4.850023e+07
2012-04-20    4.769358e+07
2012-04-27    4.580799e+07
2012-05-04    4.622683e+07
2012-05-11    4.540891e+07
2012-05-18    4.604039e+07
2012-05-25    4.562105e+07
2012-06-01    4.619086e+07
2012-06-08    4.594213e+07
2012-06-15    4.639705e+07
2012-06-22    4.623466e+07
2012-06-29    4.658250e+07
Freq: W-FRI, Name: predicted_mean, dtype: float64
```

Plot:

In [254…

```
plt.figure(figsize=(10,5))
plt.plot(sales, label="Historical")
plt.plot(future, label="Future Forecast")
plt.legend()
plt.show()
```



Forecasting using SARIMA

In [255…

```
#pip install pmdarima
```

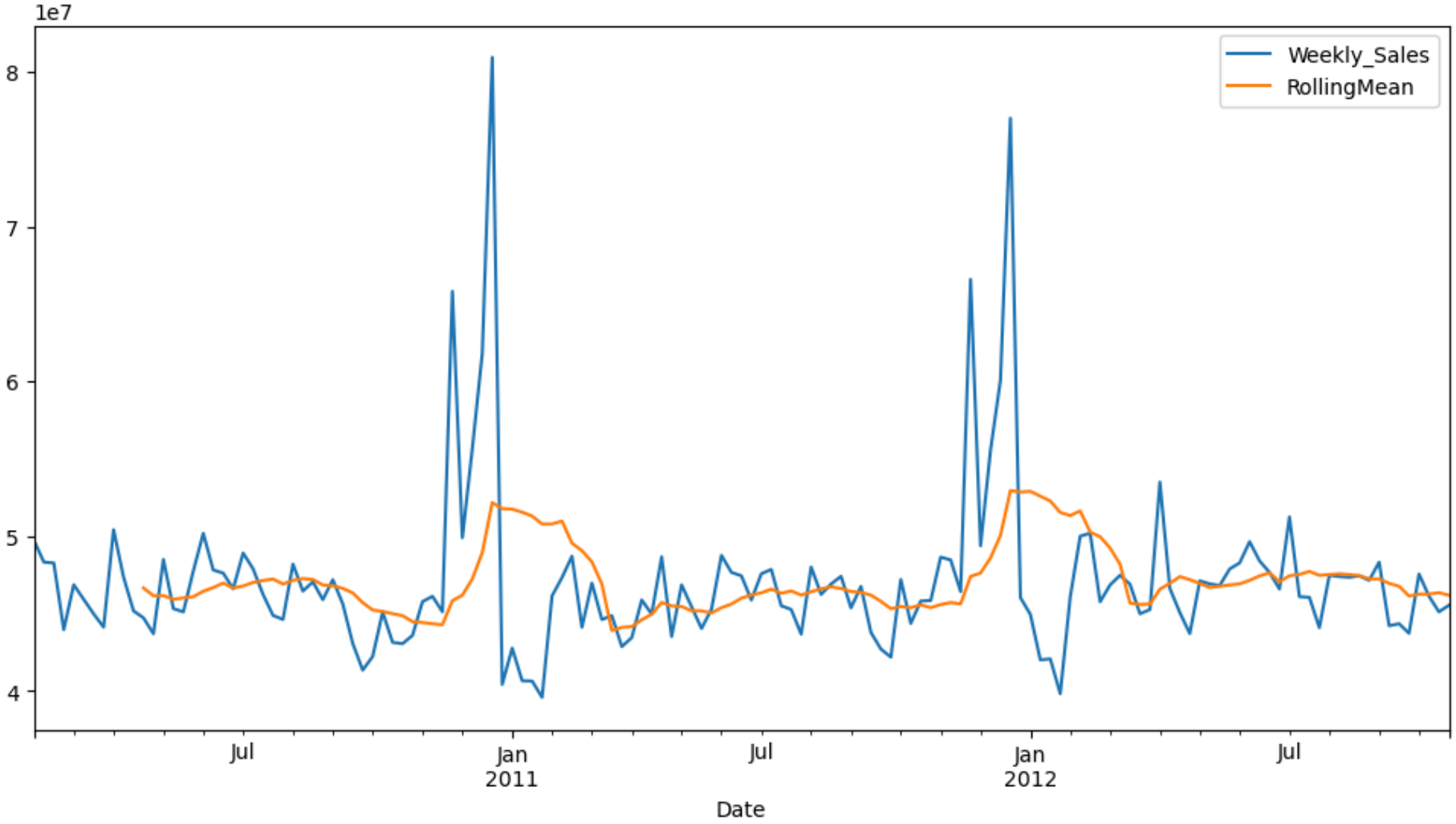
Rolling mean visualization:

In [256…

```
sales['RollingMean'] = sales['Weekly_Sales'].rolling(window=12).mean()
sales[['Weekly_Sales','RollingMean']].plot(figsize=(12,6))
```

Out[256…

```
<Axes: xlabel='Date'>
```



Feature Engineering (Lag Features)

In [257…

```
sales['lag_1'] = sales['Weekly_Sales'].shift(1)
sales['lag_2'] = sales['Weekly_Sales'].shift(2)
```

```
sales['lag_3'] = sales['Weekly_Sales'].shift(3)
sales.dropna(inplace=True)
```

auto_arima

In [258…

```
from pmdarima import auto_arima
model = auto_arima(train,
                   seasonal=True,
                   m=12,
                   trace=True)
```

```
Performing stepwise search to minimize aic
 ARIMA(2,0,2)(1,0,1)[12] intercept   : AIC=inf, Time=1.42 sec
 ARIMA(0,0,0)(0,0,0)[12] intercept   : AIC=3886.294, Time=0.03 sec
 ARIMA(1,0,0)(1,0,0)[12] intercept   : AIC=3876.056, Time=0.31 sec
 ARIMA(0,0,1)(0,0,1)[12] intercept   : AIC=3879.931, Time=0.10 sec
 ARIMA(0,0,0)(0,0,0)[12]             : AIC=4356.186, Time=0.02 sec
 ARIMA(1,0,0)(0,0,0)[12] intercept   : AIC=3874.370, Time=0.04 sec
 ARIMA(1,0,0)(0,0,1)[12] intercept   : AIC=3875.879, Time=0.11 sec
 ARIMA(1,0,0)(1,0,1)[12] intercept   : AIC=inf, Time=1.06 sec
 ARIMA(2,0,0)(0,0,0)[12] intercept   : AIC=3874.673, Time=0.06 sec
 ARIMA(1,0,1)(0,0,0)[12] intercept   : AIC=3875.637, Time=0.06 sec
 ARIMA(0,0,1)(0,0,0)[12] intercept   : AIC=3878.644, Time=0.05 sec
 ARIMA(2,0,1)(0,0,0)[12] intercept   : AIC=3876.666, Time=0.14 sec
 ARIMA(1,0,0)(0,0,0)[12]             : AIC=inf, Time=0.03 sec

Best model:  ARIMA(1,0,0)(0,0,0)[12] intercept
Total fit time: 3.450 seconds
```

SARIMAX

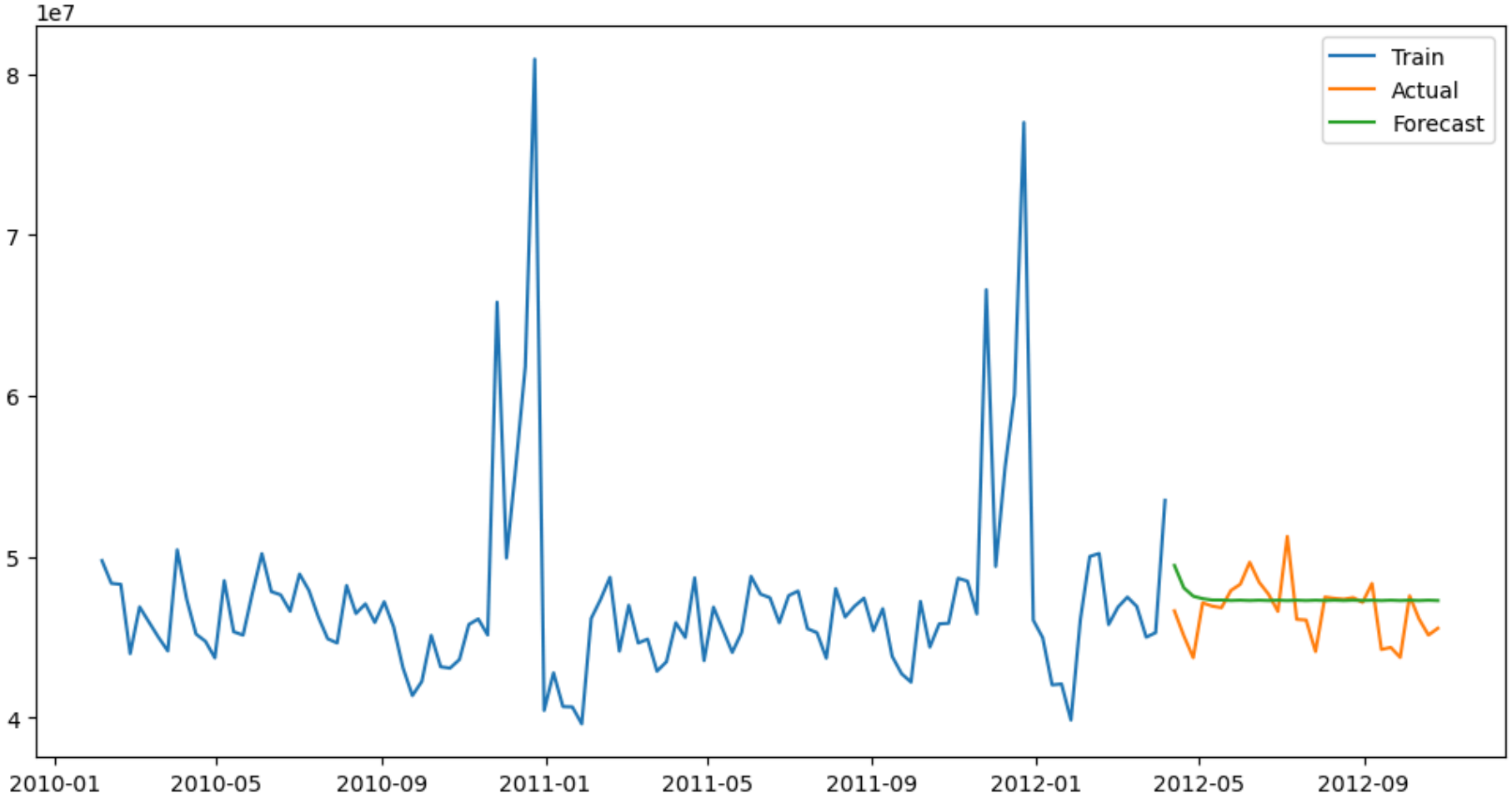
In [259…

```
model = SARIMAX(train['Weekly_Sales'],
                order=(3,0,2),
                seasonal_order=(0,0,0,12))
result = model.fit()
forecast = result.forecast(steps=len(test))
```

FORECAST

In [260…

```
plt.figure(figsize=(12,6))
plt.plot(train.index, train['Weekly_Sales'], label='Train')
plt.plot(test.index, test['Weekly_Sales'], label='Actual')
plt.plot(test.index, forecast, label='Forecast')
plt.legend()
plt.show()
```



Model Evaluation

In [261…

```
mae = mean_absolute_error(test, forecast)
rmse = np.sqrt(mean_squared_error(test, forecast))
y_true = test['Weekly_Sales']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 3.228185738954301
MAE: 1485274.2502554958
RMSE: 1951094.8243009693
```

Forecast Future Sales

In [262…

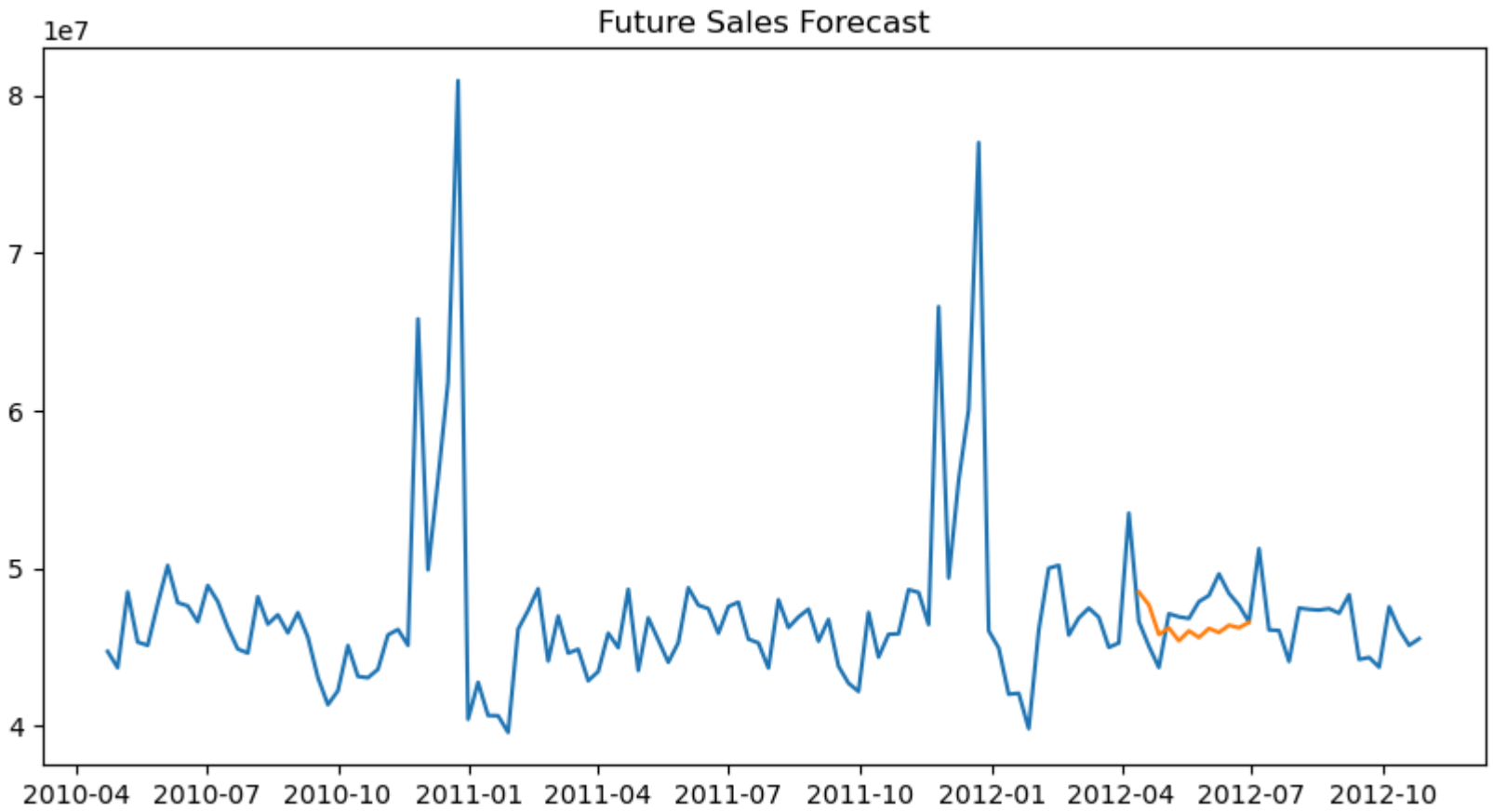
```
future_sales = model_fit.forecast(steps=12)
print(future)
```

```
2012-04-13    4.850023e+07
2012-04-20    4.769358e+07
2012-04-27    4.580799e+07
2012-05-04    4.622683e+07
2012-05-11    4.540891e+07
2012-05-18    4.604039e+07
2012-05-25    4.562105e+07
2012-06-01    4.619086e+07
2012-06-08    4.594213e+07
2012-06-15    4.639705e+07
2012-06-22    4.623466e+07
2012-06-29    4.658250e+07
Freq: W-FRI, Name: predicted_mean, dtype: float64
```

In [263…

```
plt.figure(figsize=(10,5))
plt.plot(sales.index, sales['Weekly_Sales'])
plt.plot(future_sales.index, future_sales)
```

```
plt.title("Future Sales Forecast")
plt.show()
```



SARIMAX modeL With exogenous variables

In [150…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
from sklearn.metrics import mean_absolute_percentage_error
from pmdarima import auto_arima
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

In [151…

```
train= pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\train.csv')
features = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Forecasting\features.csv")
stores = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Forecasting\stores.csv")
```

In [152…

```
import numpy as np
percentiles = np.arange(0, 1.05, 0.05)
distribution = train["Weekly_Sales"].quantile(percentiles)
print(distribution)
```

```
0.00     -4988.9400
0.05        59.9745
0.10       291.0970
0.15       727.3000
0.20      1340.9800
0.25      2079.6500
0.30      2913.3810
0.35      3843.1830
0.40      4887.9600
0.45      6152.9025
0.50      7612.0300
0.55      9271.7585
0.60     11274.6320
0.65     13679.9195
0.70     16619.3250
0.75     20205.8525
0.80     25217.6120
0.85     32481.0620
0.90     42845.6730
0.95     61201.9510
1.00    693099.3600
Name: Weekly_Sales, dtype: float64
```

In [153…

```
train = train[train["Weekly_Sales"] > 100]
```

Merge datasets

In [154…

```
df = train.merge(features, on=["Store", "Date", "IsHoliday"])
df = df.merge(stores, on="Store")
df['Date'] = pd.to_datetime(df['Date'])
df = df.sort_values('Date')
df.head()
```

Out[154…

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	Mar
0	1	1	2010-02-05	24924.50	False	42.31	2.572	NaN	NaN	NaN	NaN	
195796	21	54	2010-02-05	310.00	False	39.05	2.572	NaN	NaN	NaN	NaN	
267304	29	9	2010-02-05	4419.73	False	24.36	2.788	NaN	NaN	NaN	NaN	
195856	21	55	2010-02-05	10500.20	False	39.05	2.572	NaN	NaN	NaN	NaN	
28344	3	97	2010-02-05	175.92	False	45.71	2.572	NaN	NaN	NaN	NaN	

Holiday Encoding

In [155…

```
df["IsHoliday"] = df["IsHoliday"].astype(int)
df.head()
```

Out[155…

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	Mar
0	1	1	2010-02-05	24924.50	0	42.31	2.572	NaN	NaN	NaN	NaN	
195796	21	54	2010-02-05	310.00	0	39.05	2.572	NaN	NaN	NaN	NaN	
267304	29	9	2010-02-05	4419.73	0	24.36	2.788	NaN	NaN	NaN	NaN	
195856	21	55	2010-02-05	10500.20	0	39.05	2.572	NaN	NaN	NaN	NaN	
28344	3	97	2010-02-05	175.92	0	45.71	2.572	NaN	NaN	NaN	NaN	

Aggregate Weekly Sales

In [156…

```
#For simplicity we forecast total weekly sales.
sales = df.groupby('Date')['Weekly_Sales'].sum().reset_index()
sales = sales.set_index('Date')
```

Type Variable Encoding

In [157…

```
df = pd.get_dummies(df, columns=["Type"], drop_first=True)
```

In [158…

```
df.head()
```

Out[158…

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	Mar
0	1	1	2010-02-05	24924.50	0	42.31	2.572	NaN	NaN	NaN	NaN	
195796	21	54	2010-02-05	310.00	0	39.05	2.572	NaN	NaN	NaN	NaN	
267304	29	9	2010-02-05	4419.73	0	24.36	2.788	NaN	NaN	NaN	NaN	
195856	21	55	2010-02-05	10500.20	0	39.05	2.572	NaN	NaN	NaN	NaN	
28344	3	97	2010-02-05	175.92	0	45.71	2.572	NaN	NaN	NaN	NaN	

Changing Type varibale to int

In [159…

```
df["Type_B"] = df["Type_B"].astype(int)
df["Type_C"] = df["Type_C"].astype(int)
```

In [160…

```
df.head()
```

Out[160…

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	Mar
0	1	1	2010-02-05	24924.50	0	42.31	2.572	NaN	NaN	NaN	NaN	
195796	21	54	2010-02-05	310.00	0	39.05	2.572	NaN	NaN	NaN	NaN	
267304	29	9	2010-02-05	4419.73	0	24.36	2.788	NaN	NaN	NaN	NaN	
195856	21	55	2010-02-05	10500.20	0	39.05	2.572	NaN	NaN	NaN	NaN	
28344	3	97	2010-02-05	175.92	0	45.71	2.572	NaN	NaN	NaN	NaN	

Add Exogenous Variables

In [161…

```
exog = df.groupby('Date')[['Temperature','Fuel_Price','CPI','Unemployment',"IsHoliday","Size","Type_B","Type_C"]].mean()
```

Train Test Split

In [162…

```
train_size = int(len(sales)*0.8)
train_sales = sales[:train_size]
test_sales = sales[train_size:]
train_exog = exog[:train_size]
test_exog = exog[train_size:]
```

Build SARIMAX Model

In [163…

```
model = SARIMAX(train_sales,
                exog=train_exog,
                order=(3,0,2),
                seasonal_order=(0,0,0,12))
results = model.fit()
print(results.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:           Weekly_Sales   No. Observations:                  114
Model:               SARIMAX(3, 0, 2)   Log Likelihood               -1918.255
Date:                Thu, 12 Mar 2026   AIC                           3864.509
Time:                        19:19:20   BIC                           3902.816
Sample:                    02-05-2010   HQIC                          3880.056
                         - 04-06-2012
Covariance Type:                  opg
==================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Temperature  -5573.8888   4.04e+04     -0.138      0.890   -8.48e+04    7.37e+04
Fuel_Price   -3.716e+06   1.48e+04   -251.307      0.000   -3.74e+06   -3.69e+06
CPI           2.383e+06   8.31e+04     28.679      0.000    2.22e+06    2.55e+06
Unemployment  1.671e+07   9780.240   1708.632      0.000    1.67e+07    1.67e+07
IsHoliday     2.759e+06   1116.737   2470.649      0.000    2.76e+06    2.76e+06
Size         -5341.1449    107.443    -49.712      0.000   -5551.729   -5130.561
Type_B        6.913e+08     29.694   2.33e+07      0.000    6.91e+08    6.91e+08
Type_C       -7.993e+07     20.788  -3.85e+06      0.000   -7.99e+07   -7.99e+07
ar.L1            0.4195      0.252      1.664      0.096      -0.075       0.914
ar.L2            0.6920      0.222      3.116      0.002       0.257       1.127
ar.L3           -0.3425      0.087     -3.941      0.000      -0.513      -0.172
ma.L1           -0.3210      0.321     -1.000      0.317      -0.950       0.308
ma.L2           -0.6774      0.278     -2.440      0.015      -1.221      -0.133
sigma2        2.437e+13   6.41e-05    3.8e+17      0.000    2.44e+13    2.44e+13
===================================================================================
Ljung-Box (L1) (Q):                   0.03   Jarque-Bera (JB):               266.09
Prob(Q):                              0.86   Prob(JB):                         0.00
Heteroskedasticity (H):               3.15   Skew:                             1.69
Prob(H) (two-sided):                  0.00   Kurtosis:                         9.68
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
[2] Covariance matrix is singular or near-singular, with condition number 4.87e+33. Standard errors may be unstable.
```

In [164…

```
forecast = results.predict(start=len(train_sales),
                           end=len(sales)-1,
                           exog=test_exog)
```

Plot Forecast vs Actual

In [165…

```
import matplotlib.pyplot as plt
plt.figure(figsize=(10,5))
plt.plot(train_sales, label="Train")
plt.plot(test_sales, label="Actual")
plt.plot(forecast, label="Forecast")
plt.legend()
plt.title("Walmart Weekly Sales Forecast")
plt.show()
```

Evaluate Forecast

In [166…

```
mae = mean_absolute_error(test_sales, forecast)
rmse = np.sqrt(mean_squared_error(test_sales, forecast))
y_true = test_sales['Weekly_Sales']
y_pred = forecast
mape = mean_absolute_percentage_error(y_true, y_pred)
print("MAPE:", mape*100)
print("MAE:", mae)
print("RMSE:", rmse)
```

```
MAPE: 5.222797123653322
MAE: 2445195.762007369
RMSE: 2865191.49492597
```

Machine learning Models

In [167…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from sklearn.metrics import mean_absolute_error, mean_squared_error
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
from sklearn.metrics import mean_absolute_percentage_error
from pmdarima import auto_arima
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

In [168…
```
train= pd.read_csv(r'C:\Users\anupa\Desktop\Python Learning\Forecasting\train.csv')
features = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Forecasting\features.csv")
stores = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Forecasting\stores.csv")
```

Keep Only Positive Sales

In [169…
```
train = train[train["Weekly_Sales"] > 100]
train = train.reset_index(drop=True)
```

Convert Date

In [170…
```
train["Date"] = pd.to_datetime(train["Date"])
features["Date"] = pd.to_datetime(features["Date"])
```

Merge Datasets

In [171…
```
df = train.merge(features, on=["Store","Date","IsHoliday"], how="left")
df = df.merge(stores, on="Store", how="left")
```

Holiday Encoding

In [172…
```
df["IsHoliday"] = df["IsHoliday"].astype(int)
```

Create Time Features and sort data

In [173…
```
df["month"] = df["Date"].dt.month
df["week"] = df["Date"].dt.isocalendar().week.astype(int)
```

```
df["year"] = df["Date"].dt.year
df = df.sort_values(["Store","Dept","Date"])
```

Create Lag Features

In [174…

```
df["lag_1"] = df.groupby(["Store","Dept"])["Weekly_Sales"].shift(1)
df["lag_2"] = df.groupby(["Store","Dept"])["Weekly_Sales"].shift(2)
df["lag_3"] = df.groupby(["Store","Dept"])["Weekly_Sales"].shift(3)
```

Create Rolling Features

In [175…

```
df["rolling_mean_4"] = df.groupby(["Store","Dept"])["Weekly_Sales"].shift(1).rolling(4).mean()
df["rolling_std_4"] = df.groupby(["Store","Dept"])["Weekly_Sales"].shift(1).rolling(4).std()
```

Encode Store Type

In [176…

```
df = pd.get_dummies(df, columns=["Type"], drop_first=True)
```

Remove Missing Values

In [177…

```
df = df.dropna()
```

Define Features

In [178…

```
features_cols = [
"Store",
"Dept",
"IsHoliday",
"Temperature",
```

```
"Fuel_Price",
"CPI",
"Unemployment",
"Size",
"month",
"week",
"year",
"lag_1",
"lag_2",
"lag_3",
"rolling_mean_4",
"rolling_std_4",
"Type_B",
"Type_C"
]

X = df[features_cols]
y = df["Weekly_Sales"]
```

Train-Test Split (Time Based)

In [179…

```
split = int(len(df) * 0.8)

X_train = X.iloc[:split]
X_test = X.iloc[split:]

y_train = y.iloc[:split]
y_test = y.iloc[split:]
```

Random Forest Model

In [180…

```
rf = RandomForestRegressor(
    n_estimators=300,
    max_depth=12,
    random_state=42
)
```

```
rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)
```

XGBoost Model

In [181…

```
xgb = XGBRegressor(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    random_state=42
)

xgb.fit(X_train, y_train)

xgb_pred = xgb.predict(X_test)
```

In [182…

```
print(y_test.describe())
```

```
count     18583.000000
mean      17125.628981
std       21791.055250
min         100.010000
25%        3096.280000
50%        8817.000000
75%       21990.920000
max      351553.980000
Name: Weekly_Sales, dtype: float64
```

Model Evaluation

In [183…

```
rf_mae = mean_absolute_error(y_test, rf_pred)
rf_rmse = np.sqrt(mean_squared_error(y_test, rf_pred))
rf_mape = mean_absolute_percentage_error(y_test, rf_pred) * 100
```

```
xgb_mae = mean_absolute_error(y_test, xgb_pred)
xgb_rmse = np.sqrt(mean_squared_error(y_test, xgb_pred))
xgb_mape = mean_absolute_percentage_error(y_test, xgb_pred) * 100

print("Random Forest")
print("MAE:", rf_mae)
print("RMSE:", rf_rmse)
print("MAPE:", rf_mape)

print("\nXGBoost")
print("MAE:", xgb_mae)
print("RMSE:", xgb_rmse)
print("MAPE:", xgb_mape)
```

```
Random Forest
MAE: 2063.5616408559854
RMSE: 5403.1800010930865
MAPE: 21.250613845346198

XGBoost
MAE: 1676.5457008510602
RMSE: 3653.9462372801117
MAPE: 23.604749468909045
```

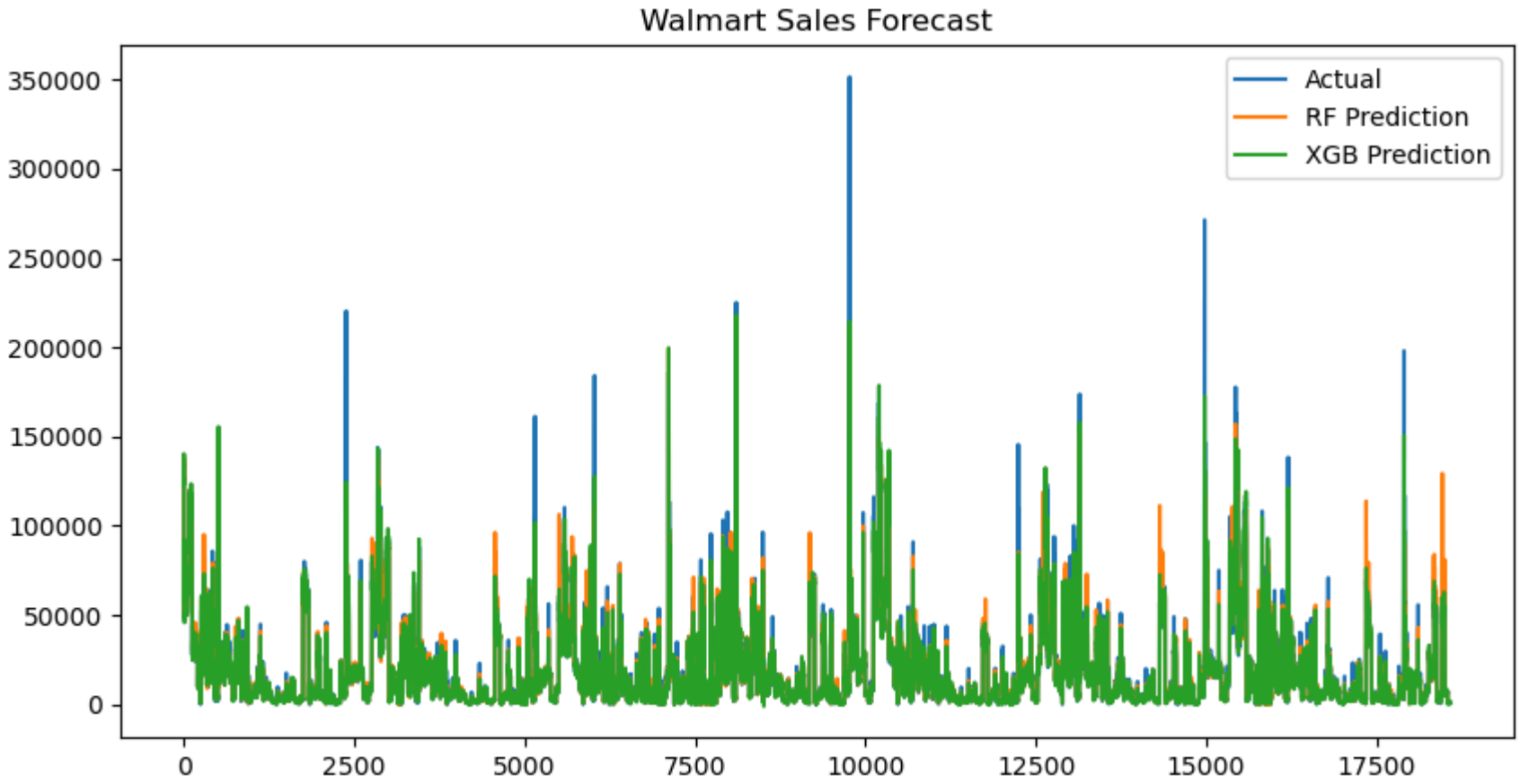
Plot Forecast

In [184…

```
plt.figure(figsize=(10,5))

plt.plot(y_test.values, label="Actual")
plt.plot(rf_pred, label="RF Prediction")
plt.plot(xgb_pred, label="XGB Prediction")

plt.legend()
plt.title("Walmart Sales Forecast")
plt.show()
```


Walmart Sales Forecast
Actual
RF Prediction
XGB Prediction
350000
300000
250000
200000
150000
100000
50000
0
0
2500
5000
7500
10000
12500
15000
17500